PROF. DR.-ING. MARK SCHUTERA

_SUPERVISED LEARNING

UNFINISHED LECTURE NOTES

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"IN MY DEFENSE, I WAS LEFT UNSUPERVISED".

DER ZAUBERLEHRLING

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Introduction

Supervised learning is limited to tasks where we have labeled data and clear objectives. Beyond this limited context of learning, we fall back to the pattern in the data itself.

We explore how pattern emerges, from other sources of information beyond explicit labels. We will see how pattern can be discovered in data, represented, leveraged and utilized. In that sense this course is about any form of learning that is not supervised learning.

Each lecture follows a consistent pattern. To start of the contents of the lecture will be motivated by a narrative, a theoretical primer, and some sort of hands-on experience to build intuition. Then we will dive into the, theory and algorithms, then reflection. Theory sessions are 45 minutes, followed by 45 minutes of coding practice to enforce reflection and deepen understanding of the concepts.

A Philosophy of Education

The beautiful risk of education is that it is an encounter. I am trying to model a way of thinking, and reasoning, then stepping back. The rest is yours.

- **Start with why.** A method understood only procedurally is a method you will misapply when it matters. Before learning how something works, understand why it was built, what it replaced, and where it fails. The mechanics then follow from necessity, not memorization.
- **Build taste alongside competence.** You need a sense of what beautiful is, you need to train your aesthetics before you can reliably produce it. This develops by studying what others have built, making it your own, then making your own, and honestly evaluating the gap between them. Taste is the compass. Skill is the engine. You need both, in that order.
- **Failure is the designed medium.** Each attempt teaches what a clean run cannot. The insight compounds across tries, not from a single correct execution. Those who never start failing will fail.
- **Work in the open.** Showing half-finished thinking, naming mistakes as they happen, is a professional skill. We practice it here. It is how knowledge actually moves between people. And it is how knowledge moves towards you, at unprecedented speed.
- **The fallacy of mediocracy.** Iteration and failure are the method, not the excuse for settling. Do not confuse velocity with quality. Whatever you learn, whatever you build: make it good.
- **Outcomes emerge. They are not delivered.** Judgment, taste, the capacity to sit with uncertainty: these are not course outputs. They are what you build from the encounter. I can create the conditions. The rest depends on what you do with them.

Part I

Geometry of Pattern

Part II

Representing Pattern

Dimensionality Reduction

Folding and Unfolding Space.

The Why

Dimensionality reduction finds those factors and discards the rest.

Real-world data $\mathbf{X} \in \mathbb{R}^{n \times d}$ is often high-dimensional: images have millions of pixels, genomes have billions of base pairs, text embeddings have thousands of dimensions. Every clustering method in Part I takes such a feature space as given and works within it. But while dimensionality comes at a cost and degrades our approaches, dimensionality carries information. The question then becomes whether a more compact representation could carry the same information.

MNIST handwritten digits , for example, live in $\mathbb{R}^{784}$ (28×28 pixels), but vary along a handful of latent factors such as shapes or geometric primitives. The intrinsic dimensionality is estimated at roughly $k \approx 12$ to 14.

Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998. doi: 10.1109/5.726791. http://yann.lecun.com/exdb/mnist/

Dimensionality has a price. In low dimensions, some points are close and others are far away; that contrast is what makes clustering work. As d grows, the contrast disappears: all pairwise distances converge toward the same value.

Computational cost grows with d, too. Distance computations scale as $O(nd)$, covariance matrices require $O(d^2)$ storage, and eigendecompositions cost $O(d^3)$. Reducing from d to k before fitting a downstream model can turn an intractable pipeline into a practical one.

Expressiveness of the feature space. Original features are often redundant or entangled. If two features $x^{(i)}$ and $x^{(j)}$ are highly correlated ($|\rho_{ij}| \approx 1$), they carry nearly the same information; a good representation replaces them with a single component that captures their shared variance.

Interpretability benefits directly. Humans usually usually do not think and reason in high-dimensional vector spaces; we prefer

low-dimensional, disentangled representations that align with our concepts of space.

In order to move from high-dimensional observations to compact, signal-carrying representations we have good reason to find ways to,

- find a mapping that retains the pattern that carries the signal.
- construct proxy dimensions that are not limited to the original features, so that hidden patterns become apparent.
- understand what different notions of preserving pattern imply, and choose the right one for a given task.

Hands-On Experience

CONSIDER the canonical six points, but now arranged so that $x^{(1)}$ varies only slightly while $x^{(2)}$ carries nearly all the spread.

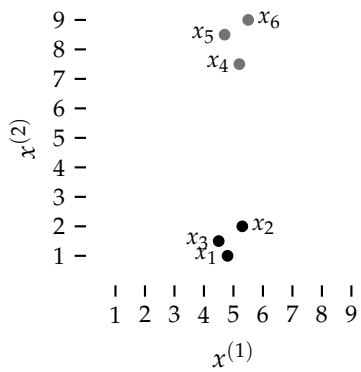


Figure 1: The canonical six points, repositioned so that $x^{(1)}$ barely varies while $x^{(2)}$ carries both the cluster separation and the within-cluster spread. Two features describe each point, but the data is essentially one-dimensional.

THE TWO CLUSTERS are clearly visible, and one feature carries almost all the signal. The first feature $x^{(1)}$ hovers around 5 for every point; dropping it loses almost nothing. The second feature $x^{(2)}$ alone separates the two groups and captures the within-cluster variation.

- If you had to keep only one of the two features, which would you choose?
- How much information would you lose?
- And could you recover the two clusters from that single feature alone?

THE METHODS IN THIS CHAPTER AUTOMATE THAT JUDGMENT OF DIMENSIONALITY REDUCTION. What is the most efficient mapping you can find for the MNIST dataset, using only geometric primitives such as circles, lines so that each class is well-separated?

HAND-CRAFTED FEATURE ENGINEERING is dimensionality reduction by another name. When a domain expert selects "age" and "income" from a database with hundreds of columns, or computes edge gradients from raw pixels, they are projecting from $\mathbb{R}^d$ to $\mathbb{R}^k$ using human genius.

A WELL-CHOSEN MAPPING $f\colon \mathbb{R}^d \to \mathbb{R}^k$ disentangles latent factors, yielding coordinates that are more discriminative for downstream tasks than the originals.

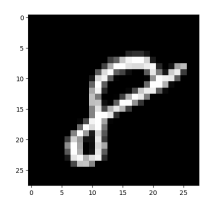


Figure 2: A single MNIST digit its meaningful variation spans few dimensions.

THE LEARNING OBJECTIVES of this lecture:

- Understand naive feature selection based on variance and its limitations.
- Understand the PCA objective and its connection to variance maximization and the singular value decomposition.
- Apply t-SNE and UMAP for visualization and avoid common misinterpretations of their output.
- Choose appropriate dimensionality reduction methods based on linearity, dataset scale, and whether the goal is preprocessing or visualization.

The Curse of Dimensionality, Revisited

THE KEY OBSERVATION is that high d is only fatal when all d dimensions carry independent information. In practice they rarely do. Features are correlated, redundant, or simply noisy. The effective degrees of freedom, the intrinsic dimensionality k, is often far smaller than d.

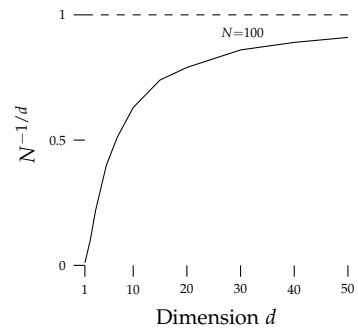


Figure 3: Mean nearest-neighbour spacing $d_{\mathrm{NN}} \propto N^{-1/d}$ for N=100 points. By d=50 every point is nearly as far from its neighbour as from the boundary. But if the data only occupies $k \ll d$ effective dimensions, the curse applies to k, not d.

REDUCING d TO k REVERSES THE CURSE. The spacing, the distance concentration, the density collapse all depend on the dimension of the space the data actually lives in. If we project from $\mathbb{R}^d$ down to $\mathbb{R}^k$, the clustering methods from Part I regain their footing: distances become discriminative again, ε-neighbourhoods fill up, and centroids sit among their members.

THE BLESSING OF DIMENSIONALITY. The Johnson-Lindenstrauss lemma shows that n points in high dimensions can be projected to $O(\log n/\epsilon^2)$ dimensions while preserving pairwise distances within $(1\pm\epsilon)$. Even random projections work.

W. B. Johnson and J. Lindenstrauss. Extensions of Lipschitz mappings into a Hilbert space. *Contemporary Mathematics*, 26:189–206, 1984. https://scholar.google.com/scholar?q=Johnson+Lindenstrauss+1984+extensions+lipschitz+mappings+hilbert+space

THE QUESTION IS HOW TO FIND THE RIGHT k DIRECTIONS. Picking original features by hand (as in variance-based selection) works when the axes happen to align with the signal. When they do not, we need methods that construct new directions from the data itself. That is what PCA, and later t-SNE and UMAP, will do.

A Naïve Baseline: Variance-Based Feature Selection

WHEN AXES ALIGN WITH THE SIGNAL consider the most straightforward way to reduce dimensions: measure the variance of each original feature independently and keep only those with the highest variance.
Given data $X \in \mathbb{R}^{n\times d}$, compute the variance along each coordinate axis:

$$\mathrm{Var}(x^{(j)}) = \frac{1}{n}\sum_{i=1}^{n}\left(x_i^{(j)} - \bar{x}^{(j)}\right)^2, \quad j = 1, \ldots, d$$

THE SIMPLEST IDEA: rank each feature by how much it varies, then drop the ones that barely move.

Sort the features by decreasing variance, keep the top k, and discard the rest. This is pure feature selection: no new axes are created, only existing ones are retained or dropped.

WHEN DOES THIS WORK? If the coordinate axes already align with the directions of meaningful variation, variance-based selection is fast, interpretable, and surprisingly effective. A near-constant feature carries almost no information, and removing it loses little.

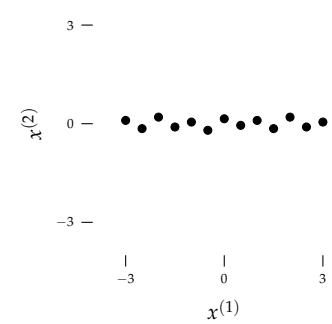


Figure 4: Data spread along $x^{(1)}$ but nearly constant in $x^{(2)}$. Dropping $x^{(2)}$ loses almost nothing.

When does it fail? When the signal does not happen to align with the axis, for example when it runs diagonally across the coordinate axes. In that case, no single original feature captures it well.

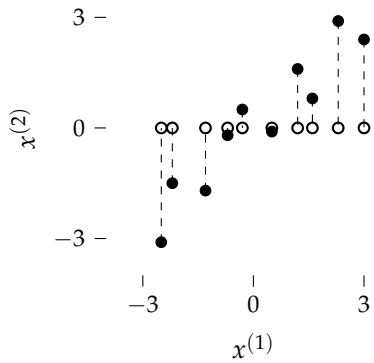


Figure 5: Naïve feature selection: dropping $x^{(2)}$ projects onto the horizontal axis. The dashed errors are large because the data varies along $y{=}x$, not along a coordinate axis. Both features have nearly equal variance, so the naïve ranking cannot decide which to keep. Notice, how in this case both features would still preserver the signal.

Principal Component Analysis

Here both features have nearly the same variance, so variance-based selection cannot decide which to keep. Dropping either axis incurs large reconstruction error. The real pattern lies along the diagonal $y{=}x$, a direction that no original feature represents. When the interesting variation runs between features, we need to construct and map to new dimensions, meaning features. That is exactly what PCA does.

The manifold hypothesis holds that high-dimensional data concentrates on a k-dimensional manifold $\mathcal{M} \subset \mathbb{R}^d$ with $k \ll d$. The goal of dimensionality reduction is to find a coordinate system for $\mathcal{M}$.

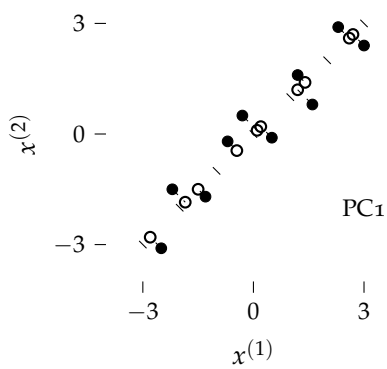


Figure 6: PCA projection onto PC1: The direction of maximum variance runs along $y{=}x$. The reconstruction errors (dashed) are tiny compared to the naïve projection.

Given centred data $X \in \mathbb{R}^{n \times d}$, where rows are samples, PCA finds a k-dimensional subspace, $k < d$, that best represents the data.

PCA finds orthogonal directions of maximum variance in the data.

K. Pearson. On lines and planes of closest fit to systems of points in space. *Philosophical Magazine*, 2(11):559–572, 1901. DOI: 10.1080/14786440109462720. https://scholar.google.com/scholar?q=Pearson+1901+lines+planes+closest+fit+systems+points; and H. Hotelling. Analysis of a complex of statistical variables into principal components. *Journal of Educational Psychology*, 24(6):417–441, 1933. DOI: 10.1037/h0071325. https://scholar.

Algorithm 1: PCA via Eigendecomposition

Require: Centred data $X \in \mathbb{R}^{n \times d}$ (each column has zero mean, so that $X^T X/n$ is the covariance matrix; if features have different scales, standardise first by dividing each column by its standard deviation), target dimension k

Ensure: Principal components $W \in \mathbb{R}^{d \times k}$, projected data $Z \in \mathbb{R}^{n \times k}$

1: Compute covariance matrix $\Sigma = X^T X/n$
2: Compute eigendecomposition $\Sigma = V \Lambda V^T$
3: $W \leftarrow$ first k columns of V (sorted by decreasing eigenvalue)
4: $Z \leftarrow XW$ ▷ Project data onto k-dimensional subspace

THE VARIANCE MAXIMIZATION VIEW. Find the direction $w_1 \in \mathbb{R}^d$ with $\|w_1\|=1$ that maximises the variance of projected data:

$$\begin{aligned} w_1 &= \arg\max_{\|w\|=1} \mathrm{Var}(Xw) \\ &= \arg\max_{\|w\|=1} w^T \Sigma\, w, \\ \Sigma &= \tfrac{1}{n} X^T X. \end{aligned}$$

LAGRANGE MULTIPLIERS. Maximising $w^T \Sigma w$ subject to $w^T w = 1$ is a constrained optimisation problem. Introduce a multiplier λ and form the Lagrangian:

$$\mathcal{L}(w, \lambda) = w^T \Sigma\, w \; - \; \lambda\, (w^T w - 1)$$

Differentiating with respect to w and setting the gradient to zero:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial w} &= 2\, \Sigma\, w \; - \; 2\, \lambda\, w \\ &\overset{!}{=} \; 0 \end{aligned}$$

which gives the eigenvalue equation:

$$\Sigma\, w = \lambda\, w$$

So w must be an eigenvector of Σ. The projected variance along any such eigenvector equals its eigenvalue:

$$\begin{aligned} w^T \Sigma\, w &= w^T (\lambda\, w) \\ &= \lambda\, w^T w \\ &= \lambda \end{aligned}$$

Maximising the variance therefore means choosing w_1 as the eigenvector with the largest eigenvalue λ_1. The second principal component is the eigenvector with the second-largest eigenvalue, orthogonal to the first, and so on. The first k principal components together define the optimal k-dimensional subspace. $k \leq d$ is a free choice.

ORTHOGONALITY of principal components follows directly: eigenvectors of a symmetric matrix corresponding to distinct eigenvalues are orthogonal.

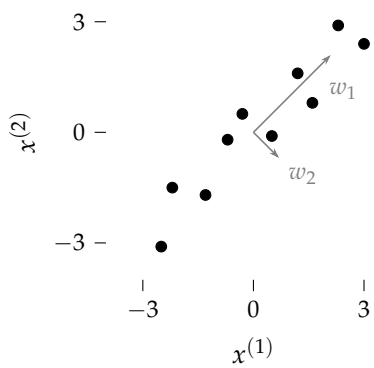


Figure 7: The two eigenvectors of Σ. w_1 points along the direction of maximum variance; w_2 is orthogonal and captures the residual spread.

COMPUTING EIGENVALUES. Eigenvalues satisfy the characteristic equation $\det(\Sigma - \lambda I) = 0$. For a 2×2 covariance matrix $\Sigma = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$:

$$\begin{aligned} \det(\Sigma - \lambda I) &= \det \begin{pmatrix} a-\lambda & b \\ b & c-\lambda \end{pmatrix} \\ &= (a-\lambda)(c-\lambda) - b^2 \\ &= \lambda^2 - (a+c)\,\lambda + (ac - b^2) \\ &\overset{!}{=} 0 \end{aligned}$$

GEOMETRIC INTUITION. Σ stretches space along its eigenvector directions. Subtracting λI weakens every direction by λ. When λ exactly matches the stretch along some eigenvector, $\Sigma - \lambda I$ collapses that direction to zero, the matrix becomes singular, and $\det = 0$. So the characteristic equation asks: at which scaling factors λ does Σ lose a dimension?

Applying the quadratic formula:

$$\begin{aligned} \lambda &= \frac{(a+c) \pm \sqrt{(a+c)^2 - 4(ac - b^2)}}{2} \\ &= \frac{(a+c) \pm \sqrt{(a-c)^2 + 4b^2}}{2} \end{aligned}$$

COMPUTING EIGENVECTORS. For each eigenvalue λ_i, solve $(\Sigma - \lambda_i I)\, w = 0$. From the first row of the 2×2 system:

$$(a - \lambda_i)\, w^{(1)} + b\, w^{(2)} = 0$$

$$w^{(2)} = -\frac{a - \lambda_i}{b}\, w^{(1)}$$

Any non-zero solution gives the direction; normalise to $\|w_i\|=1$ to obtain the principal component.

A SIMPLE DIAGNOSTIC reveals whether the data is compressible. Inspect the eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_d$ of Σ. If most are near zero, the data varies in only a few directions; the remaining dimensions add volume but no information.

The scree plot (Figure 8) visualises exactly this: a steep drop tells you how many components matter.

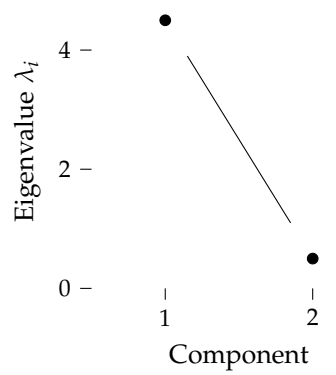


Figure 8: Scree plot for the two-component example (λ_1=4.5, λ_2=0.5). The sharp drop confirms that one direction captures most of the variance.

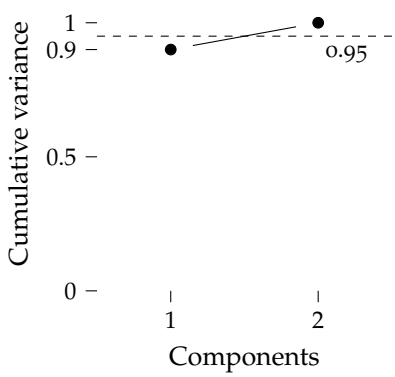


Figure 9: Cumulative explained variance: the fraction of total variance captured by the first k components is $\sum_{i=1}^{k} \lambda_i / \sum_{i=1}^{d} \lambda_i$. Here PC1 alone explains 90%; both components are needed to exceed the rule of thumb 95% threshold (dashed).

Reconstruction Error

SO FAR WE DERIVED PCA FROM VARIANCE MAXIMISATION: find the directions along which the projected data spreads the most. There is a dual view that asks the opposite question: if we compress from d dimensions to k and then decompress back to d, how much signal do we lose?

GIVEN CENTRED DATA $X \in \mathbb{R}^{n \times d}$ and an orthonormal basis $W \in \mathbb{R}^{d \times k}$ ($W^T W = I_k$), each point x_i is projected to $z_i = W^T x_i \in \mathbb{R}^k$ and reconstructed as $\hat{x}_i = W z_i = W W^T x_i$. The reconstruction error is the squared distance between each point and its reconstruction:

$$\begin{aligned}\mathcal{L}(R &= \sum_{i=1}^{n} \|x_i - \hat{x}_i\|^2 \\ &= \sum_{i=1}^{n} \|x_i - W W^T x_i\|^2\end{aligned}$$

THE DASHED ERROR segments are short because PC1 captures the dominant direction of variation, making x and its reconstruction similar.

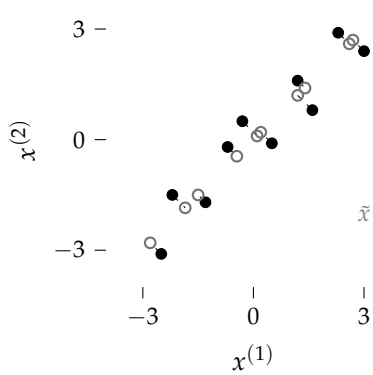


Figure 10: Reconstruction error of PCA with k=1. Each point (filled) is projected onto PC1 (the line y=x) to get its reconstruction (open circle). The dashed segments are the reconstruction errors: the perpendicular distances that PCA minimises. Compare with Figure 5, where projection onto a coordinate axis leaves much larger errors.

FOR $d > 2$, closed-form eigendecompositions do not exist. In practice, use the SVD: $X = U \Sigma V^T$, where the columns of V are the eigenvectors of $X^T X$ with $\lambda_i = \sigma_i^2 / n$. This avoids forming $X^T X$ explicitly and is numerically stable.

Independent Component Analysis

PCA DECORRELATES but does not separate sources. Uncorrelated ($\mathrm{Cov}(s_i, s_j)$=0) is weaker than independent ($p(s_i, s_j) = p(s_i)\, p(s_j)$). Given a linear mixture $X = AS$ of independent signals S, ICA recovers $S = A^{-1} X$ by maximising non-Gaussianity of the components.

t-SNE[1]

[1] L. van der Maaten and G. Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605, 2008. https://scholar.google.com/scholar?q=van+der+Maaten+Hinton+2008+visualizing+data+t-SNE

T-SNE. stands for t-distributed Stochastic Neighbour Embedding. It is optimised for one-, two- or three-dimensional visualization. t-SNE preserves local pattern; nearby points in high dimensions stay nearby in the embedding. It does this in four steps.

Algorithm 2: t-SNE. All points are updated simultaneously (batch gradient descent). The KL objective is non-convex, so there is no convergence guarantee; different random initialisations can yield different layouts.

Require: High-dimensional data X, perplexity, target dimension k

1: Compute pairwise similarities in high-D (Gaussian kernel, maps distances to $(0, 1]$):

$$p_{j|i} = \frac{\exp(-\|x_i - x_j\|^2/2\sigma_i^2)}{\sum_{l \neq i} \exp(-\|x_i - x_l\|^2/2\sigma_i^2)}$$

Symmetrize: $p_{ij} = (p_{j|i} + p_{i|j})/2n$

2: Initialize low-D embedding Y

3: **repeat**

4: Compute low-D similarities with Student-t kernel:

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}}$$

5: Gradient step: minimise

$$\mathrm{KL}(P\|Q) = \sum_{ij} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

6: **until** convergence

PCA AND ICA ARE LINEAR. No rotation of the axes can untangle classes that live on a curved manifold. When the goal is visualization rather than preprocessing, we need methods that can bend the coordinate system to follow the data. This is what t-SNE and UMAP provide.

PERPLEXITY controls the effective number of neighbours each point considers. It sets the bandwidth σ_i of the Gaussian kernel. Typical values: 5 to 50. Higher perplexity means more global pattern is considered.

PERPLEXITY is defined as $2^{H(P_i)}$ where H is the Shannon entropy of the conditional distribution over neighbours. Think of it as the effective number of neighbours.

THE GRADIENT of $\mathrm{KL}(P\|Q)$ with respect to each embedding coordinate is

$$\frac{\partial\, \mathrm{KL}}{\partial\, y_i} = 4 \sum_{j \neq i} (p_{ij} - q_{ij})\,(y_i - y_j)\,\left(1 + \|y_i - y_j\|^2\right)^{-1}$$

THE KL DIVERGENCE $\mathrm{KL}(P\|Q) = \sum_{ij} p_{ij} \log(p_{ij}/q_{ij})$ measures how much the low-dimensional distribution Q diverges from the high-dimensional distribution P. It is zero when $Q = P$ and asymmetric: misrepresenting a large p_{ij} with a small q_{ij} is penalised heavily, so t-SNE prioritises preserving local neighbourhoods.

Each summand acts as a pairwise force: where $p_{ij} > q_{ij}$ the force is attractive, where $p_{ij} < q_{ij}$ it is repulsive. The heavy tails of the Student-t kernel give distant pairs in Y a larger q_{ij} than a Gaussian would, so the repulsive force saturates. This prevents the crowding

problem where all points collapse into the centre of the map. Gradient descent moves the embedding until local neighbourhoods in Y match those in X.

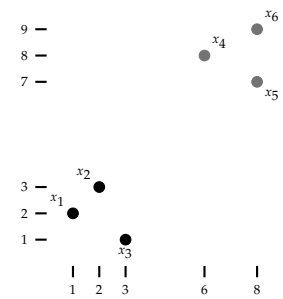
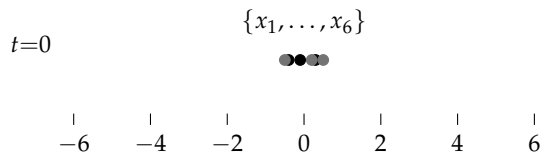


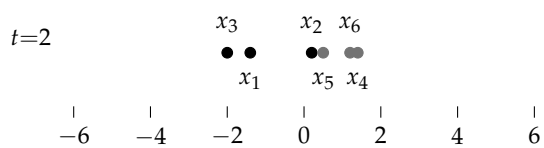
Figure 11: The canonical six points (again)). Cluster A: $x_1{=}(1,2)$, $x_2{=}(2,3)$, $x_3{=}(3,1)$. Cluster B: $x_4{=}(6,8)$, $x_5{=}(8,7)$, $x_6{=}(8,9)$.

Trace the four steps on our canonical six points, mapping from $\mathbb{R}^2$ to $\mathbb{R}^1$ with perplexity=2.

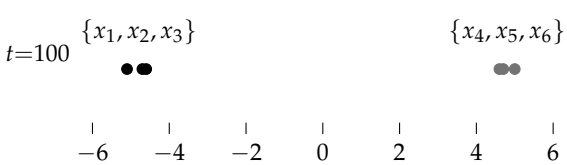
Step 1: Initialise the embedding. Each y_i is drawn independently from $\mathcal{N}(0, 10^{-4})$. At $t{=}0$ the six points land in near-random order: cluster A (black) and cluster B (grey) are completely intermixed.



Step 2: Compute high-dimensional similarities and begin gradient descent. Here perplexity $=$ 2. Same-cluster pairs get large similarities, cross-cluster pairs near zero; the resulting matrix P is computed once and stays fixed as the target. Gradient descent then pulls high-similarity neighbours together and pushes low-similarity points apart. By $t{=}2$ most points drift toward their cluster partners, but x_2; initialised near x_4 (see $t{=}0$), needed a few steps for its atractive forces to move through the repulsive forces of the other cluster's points.



Step 3: Convergence. At every iteration, q_{ij} is recomputed from the current embedding distances. The forces balance and the embedding stabilises. Within each cluster the ordering mirrors similarity: x_1/x_2 (highest $p_{12}{=}0.098$) land closest in A, x_5/x_6 ($p_{56}{=}0.089$) in B.



Interpreting t-SNE. Cluster sizes and inter-cluster distances are meaningless. t-SNE equalises local densities and preserves only neighbourhood structure. Results vary with the random seed; always run multiple times. Use t-SNE for visualization only, never as preprocessing for downstream models.

UMAP

UMAP stands for Uniform Manifold Approximation and Projection . It has theoretical foundations in Riemannian geometry and algebraic topology.

L. McInnes, J. Healy, and J. Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018. https://arxiv.org/abs/1802.03426

UMAP BRINGS THREE PRACTICAL STRENGTHS.

SCALE. UMAP operates on a sparse k-nearest-neighbour graph instead of all n^2 pairs, giving $O(n \log n)$ complexity in practice, large enough for million-point datasets.

GLOBAL STRUCTURE. UMAP optimises a cross-entropy loss that penalises both pulling true neighbours apart *and* pushing true non-neighbours together, so the embedding preserves large-scale arrangement alongside local clusters.

REPRODUCIBILITY. UMAP initialises the embedding with a spectral embedding of the graph Laplacian, so repeated runs on the same data converge to similar layouts rather than producing arbitrary rotations each time.

Algorithm 3: UMAP. Like t-SNE, all points are updated simultaneously and the objective is non-convex. The spectral initialisation makes results more reproducible than t-SNE's random start, but does not guarantee a global optimum. Key parameters: `n_neighbors` (typically 15–200) controls local versus global structure, analogous to perplexity in t-SNE; `min_dist` (typically 0.1–0.5) controls how tightly points cluster in the embedding.

Require: High-dimensional data X, number of neighbours k, target dimension d, minimum distance δ

1: Build a weighted k-nearest-neighbour graph. For each x_i, set $\rho_i = \min_{j \in \mathrm{NN}(i)} \|x_i - x_j\|$ and choose σ_i so that

$$\sum_{j \in \mathrm{NN}(i)} \exp\bigl(-(\|x_i - x_j\| - \rho_i)/\sigma_i\bigr) = \log_2 k$$

2: Compute directed edge weights:

$$w_{j|i} = \exp\bigl(-\max(0,\ \|x_i - x_j\| - \rho_i)/\sigma_i\bigr)$$

Symmetrise: $w_{ij} = w_{j|i} + w_{i|j} - w_{j|i} \cdot w_{i|j}$

3: Initialise low-D coordinates Y via spectral embedding

4: **repeat**

5: Compute low-D weights:

$$v_{ij} = \bigl(1 + a\,\|y_i - y_j\|^{2b}\bigr)^{-1}$$

6: Gradient step: minimise

$$\mathcal{L} = \sum_{ij} \Bigl[w_{ij} \log \frac{w_{ij}}{v_{ij}} + (1 - w_{ij}) \log \frac{1 - w_{ij}}{1 - v_{ij}} \Bigr]$$

7: **until** convergence

Examples & Exercises

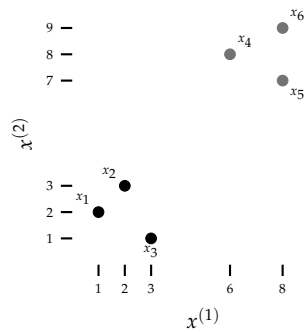


Figure 12: The canonical six points from Chapters 3 and 4. Cluster A: $x_1{=}(1,2)$, $x_2{=}(2,3)$, $x_3{=}(3,1)$. Cluster B: $x_4{=}(6,8)$, $x_5{=}(8,7)$, $x_6{=}(8,9)$.

Computing by hand builds the geometric intuition that plotting the output never can. Step through the arithmetic slowly.

Given the canonical six points $x_1{=}(1,2)$, $x_2{=}(2,3)$, $x_3{=}(3,1)$, $x_4{=}(6,8)$, $x_5{=}(8,7)$, $x_6{=}(8,9)$, centre the data, compute the covariance matrix Σ, find its eigenvalues and eigenvectors, and determine what fraction of variance PC1 explains.

Centering. PCA requires zero-mean data. Compute the sample mean:

$$\begin{aligned}\bar{x} &= \frac{1}{6}\sum_{i=1}^{6} x_i \\ &= \left(\frac{28}{6},\ \frac{30}{6}\right) \\ &= \left(\frac{14}{3},\ 5\right)\end{aligned}$$

Subtract $\bar{x}$ from each point. Here is the full calculation for x_1:

$$\begin{aligned}\tilde{x}_1 &= x_1 - \bar{x} \\ &= \left(1 - \tfrac{14}{3},\ 2 - 5\right) \\ &= \left(-\tfrac{11}{3},\ -3\right)\end{aligned}$$

Compute the remaining five yourself, then verify against the table below.

Point	**Original**	**Centred $\tilde{x}_i$**
x_1	(1, 2)	$(-11/3,\ -3)$
x_2	(2, 3)	$(-8/3,\ -2)$
x_3	(3, 1)	$(-5/3,\ -4)$
x_4	(6, 8)	$(4/3,\ 3)$
x_5	(8, 7)	$(10/3,\ 2)$
x_6	(8, 9)	$(10/3,\ 4)$

Table 1: Centred data. The second coordinate centres cleanly ($\bar{x}^{(2)}{=}5$); the first introduces thirds ($\bar{x}^{(1)}{=}14/3$).

THE COVARIANCE MATRIX. $\Sigma = \tilde{X}^T \tilde{X}/n$. Here is the full calculation for Σ_{11}:

$$\begin{aligned}
\Sigma_{11} &= \frac{1}{6}\sum_{i=1}^{6}\left(\tilde{x}_i^{(1)}\right)^2 \\
&= \frac{1}{6}\left[\frac{121}{9} + \frac{64}{9} + \frac{25}{9} + \frac{16}{9} + \frac{100}{9} + \frac{100}{9}\right] \\
&= \frac{1}{6} \cdot \frac{426}{9} \\
&= \frac{71}{9} \\
&\approx 7.89
\end{aligned}$$

Compute Σ_{12} and Σ_{22} yourself. The full matrix:

$$\Sigma = \begin{pmatrix} 71/9 & 47/6 \\ 47/6 & 29/3 \end{pmatrix} \approx \begin{pmatrix} 7.89 & 7.83 \\ 7.83 & 9.67 \end{pmatrix}$$

$\Sigma_{12} = 47/6 \approx 7.83$: both features are strongly positively correlated, as expected from the scatter plot where both coordinates increase together from cluster A to cluster B.

EIGENVALUES. Apply the formula from the theory section with $a = 71/9$, $b = 47/6$, $c = 29/3$:

$$\begin{aligned}
a + c &= \frac{71}{9} + \frac{29}{3} \\
&= \frac{158}{9} \\
&\approx 17.56
\end{aligned}$$

$$\begin{aligned}
(a-c)^2 + 4b^2 &= \left(\frac{-16}{9}\right)^2 + 4\left(\frac{47}{6}\right)^2 \\
&= \frac{256}{81} + \frac{2209}{9} \\
&= \frac{20137}{81} \\
&\approx 248.6
\end{aligned}$$

So:

$$\begin{aligned}
\lambda &= \frac{17.56 \pm \sqrt{248.6}}{2} \\
&\approx \frac{17.56 \pm 15.77}{2}
\end{aligned}$$

$\lambda_1 \approx 16.66$ and $\lambda_2 \approx 0.89$.

EIGENVECTOR FOR $\lambda_1 \approx 16.66$. Solve $(a - \lambda_1)\, w^{(1)} + b\, w^{(2)} = 0$:

$$\begin{aligned}
(7.89 - 16.66)\, w^{(1)} + 7.83\, w^{(2)} &= 0 \\
-8.77\, w^{(1)} + 7.83\, w^{(2)} &= 0 \\
w^{(2)} &= \frac{8.77}{7.83}\, w^{(1)} \\
&\approx 1.12\, w^{(1)}
\end{aligned}$$

$$\begin{aligned}
\|w_1\| &= \sqrt{1^2 + 1.12^2} \\
&= \sqrt{2.25} \\
&= 1.50, \\
w_1 &\approx \frac{1}{1.50} \begin{pmatrix} 1 \\ 1.12 \end{pmatrix} \\
&\approx \begin{pmatrix} 0.67 \\ 0.75 \end{pmatrix}.
\end{aligned}$$

EIGENVECTOR FOR $\lambda_2 \approx 0.89$. Solve $(a - \lambda_2)\, w^{(1)} + b\, w^{(2)} = 0$:

$$\begin{aligned}
7.00\, w^{(1)} + 7.83\, w^{(2)} &= 0 \\
w^{(2)} &\approx -0.89\, w^{(1)}
\end{aligned}$$

Normalising: $w_2 \approx \begin{pmatrix} 0.75 \\ -0.67 \end{pmatrix}$.

Verify orthogonality: $w_1^T w_2 = (0.67)(0.75) + (0.75)(-0.67) = 0$. ✓

EXPLAINED VARIANCE BY PC1:

$$\begin{aligned}
\frac{\lambda_1}{\lambda_1 + \lambda_2} &\approx \frac{16.66}{17.56} \\
&\approx 95\%
\end{aligned}$$

GEOMETRIC INTERPRETATION. PC1 points roughly along the direction from cluster A (bottom left) to cluster B (top right). The two clusters that are visible in two dimensions can be separated using a single principal component.

PROJECT AND RECONSTRUCT. Using the eigenvectors from the previous exercise, project each of the six centred points onto PC1, reconstruct them in the original space, and compute the reconstruction error.

PROJECTION ONTO PC1. For each centred point $\tilde{x}_i$, the coordinate on PC1 is $z_i = \tilde{x}_i^T w_1$. Here is the full calculation for x_1:

$$\begin{aligned} z_1 &= \tilde{x}_1^T w_1 \\ &= (-3.67)(0.67) + (-3)(0.75) \\ &= -2.46 - 2.25 \\ &= -4.71 \end{aligned}$$

Compute z_2 through z_6 yourself, then verify against the table below.

RECONSTRUCTION. Each projected point maps back to 2D via $\hat{x}_i = z_i \cdot w_1$. For x_1:

$$\begin{aligned} \hat{x}_1 &= -4.71 \cdot \begin{pmatrix} 0.67 \\ 0.75 \end{pmatrix} \\ &= \begin{pmatrix} -3.16 \\ -3.53 \end{pmatrix} \end{aligned}$$

RECONSTRUCTION ERROR:

$$\begin{aligned} \|\tilde{x}_1 - \hat{x}_1\|^2 &= (-3.67 + 3.16)^2 + (-3.00 + 3.53)^2 \\ &= (-0.51)^2 + (0.53)^2 \\ &= 0.26 + 0.28 \\ &= 0.54 \end{aligned}$$

Point	**Centred** $\tilde{x}_i$	z_i	**Reconstructed** $\hat{x}_i$	$\|\tilde{x}_i - \hat{x}_i\|^2$
x_1	$(-3.67, -3)$	-4.71	$(-3.16, -3.53)$	0.54
x_2	$(-2.67, -2)$	-3.29	$(-2.20, -2.47)$	0.44
x_3	$(-1.67, -4)$	-4.12	$(-2.76, -3.09)$	2.02
x_4	$(1.33, 3)$	3.14	$(2.10, 2.36)$	1.01
x_5	$(3.33, 2)$	3.73	$(2.50, 2.80)$	1.33
x_6	$(3.33, 4)$	5.23	$(3.50, 3.92)$	0.04
			Total	**5.38**

Table 2: PCA projection onto PC1 and reconstruction for the canonical six points. The total error $\approx n \cdot \lambda_2 = 6 \times 0.89 = 5.34$ (the small difference is rounding in intermediate steps). Point x_3 has the largest error because it sits furthest from the PC1 direction; x_6 has almost none because it lies nearly on PC1.

NAÏVE FEATURE SELECTION VS PCA. Using the same six centred points, compare the reconstruction error of naïve feature selection with PCA.

DROP $x^{(2)}$ (KEEP $x^{(1)}$). The reconstruction is $\hat{x}_i = (\tilde{x}_i^{(1)}, 0)$. For x_1:

$$\begin{aligned}\|\tilde{x}_1 - \hat{x}_1\|^2 &= (-3.67 - (-3.67))^2 + (-3 - 0)^2 \\ &= 0 + 9 \\ &= 9\end{aligned}$$

DROP $x^{(1)}$ (KEEP $x^{(2)}$). The reconstruction is $\hat{x}_i = (0, \tilde{x}_i^{(2)})$. For x_1:

$$\begin{aligned}\|\tilde{x}_1 - \hat{x}_1\|^2 &= (-3.67 - 0)^2 + (-3 - (-3))^2 \\ &= 13.47 + 0 \\ &= 13.47\end{aligned}$$

Compute the remaining errors yourself, then verify against the table.

Point	**Drop $x^{(2)}$**	**Drop $x^{(1)}$**	**PCA (PC1)**
x_1	9.00	13.44	0.54
x_2	4.00	7.11	0.44
x_3	16.00	2.78	2.02
x_4	9.00	1.78	1.01
x_5	4.00	11.11	1.33
x_6	16.00	11.11	0.04
Total	**58.00**	**47.33**	**5.38**

Table 3: Reconstruction error comparison. The naïve variance ranking correctly picks $x^{(2)}$ ($\mathrm{Var}(x^{(2)}) \approx 9.67 > \mathrm{Var}(x^{(1)}) \approx 7.89$), yet PCA still reduces the error by a factor of ≈ 9.

The naïve approach gets the ranking right: $x^{(2)}$ does carry more variance than $x^{(1)}$. But neither coordinate axis aligns with the direction the data actually varies along. PCA constructs that direction from the data and achieves roughly nine times less error.

CODE is provided as a Python notebook. Use it as a starting point, break things, and observe what changes.

THE CODE EXCERCISES LOAD 1 000 handwritten digits from MNIST: 784 pixels each, ten classes. In code, you will apply three preprocessing stages: discard uninformative pixels via a variance threshold, centre the data by subtracting the per-feature mean (without centring, PC1 aligns with the mean intensity rather than with the dominant direction of variation), and eigendecompose the covariance matrix to project onto the top two components. Then compare the PCA projection with t-SNE (sweeping perplexity) and UMAP (sweeping neighbourhood size).

WHAT TO OBSERVE. PCA preserves global geometry: The ten digit classes form broad, overlapping regions whose distances reflect the original space. Because PCA is linear, it cannot untangle classes that

live on a curved manifold; overlap in the 2D plot is a direct consequence. t-SNE compresses each class into a tight ball, but distances between balls are arbitrary and change with the random seed. UMAP produces similarly tight clusters while retaining more of the global arrangement. Both nonlinear methods are sensitive to their main hyperparameter; the sweep makes this visible.

Self-Reflection and Recap

SELF-REFLECTION questions to guide your thinking:

- What does it mean for data to have low intrinsic dimensionality?
- Why does PCA correspond to an eigenvector decomposition of the covariance matrix?
- What is the difference between an uncorrelated and an independent component?
- Why does t-SNE use a Student-t kernel in the low-dimensional space but a Gaussian in the high-dimensional space?
- When would you prefer UMAP over t-SNE, and when would you use PCA instead of either?
- What does the perplexity parameter actually control, and what happens when it is too low or too high?
- Why should you never use t-SNE or UMAP embeddings as features for a downstream classifier?
- How does the curse of dimensionality motivate dimensionality reduction before running K-Means?

RECAP of Key Concepts:

- PCA finds the orthogonal directions of maximum variance via eigendecomposition of the covariance matrix; SVD is the numerically preferred implementation.
- The explained-variance ratio guides the choice of k; retain enough components to exceed a threshold such as 95%.
- t-SNE and UMAP preserve local neighbourhood patterns for visualization; cluster sizes and between-cluster distances in their output are not interpretable.

- ICA finds statistically independent sources by maximising non-Gaussianity, going beyond the uncorrelated directions found by PCA.
- No non-linear method should be used as a preprocessing step for downstream learning; use PCA or SVD for that purpose.

PCA AND ICA CAN ONLY ROTATE AND SCALE AXES. When the underlying structure is nonlinear, no linear map can recover it. t-SNE and UMAP solve this for visualization, but neither learns a parametric encoder that can be applied to new data or reused as a preprocessing step.

THE FUNDAMENTAL LIMIT of PCA and ICA is linearity; they cannot capture curved structure. t-SNE and UMAP learn nonlinear embeddings, but are designed for visualization, not for learning a reusable mapping.

AUTOENCODERS learn both the compression and the reconstruction from data. The encoder is a neural network that maps x_i to a low-dimensional code z_i; the decoder maps z_i back to $\hat{x}_i$. Variational Autoencoders go one step further: the encoder outputs not a single point but a mean μ and a variance σ^2, turning the code into a distribution from which new data can be sampled.

TEASER. If we replace the fixed eigenvector projection with a learned, nonlinear encoder, can we get a compact representation that generalises to unseen inputs?

FEEDBACK

Variational Inference

EMBEDDINGS AND LATENT FEATURE SPACES.

The Why

PCA finds linear projections; t-SNE and UMAP learn nonlinear embeddings for visualization. Neither learns a parametric encoder that generalises to unseen inputs or generates new data. Autoencoders close the first gap: a neural network compresses x to a low-dimensional code z and a second network reconstructs $\hat{x}$ from z. But the resulting latent space may be irregular, with holes where no training data maps. Sampling a random point from that space and decoding it may produce nonsense.

AUTOENCODERS COMPRESS, but VAEs generate and quantify uncertainty in a single framework.

VARIATIONAL AUTOENCODERS[2] answer three questions at once by replacing the deterministic bottleneck with a probabilistic one. Instead of encoding each input to a single point, the encoder outputs a distribution. The decoder then samples from that distribution, and a regularisation term ensures the latent space is well-structured.

[2] D. P. Kingma and M. Welling. Auto-encoding variational Bayes. *arXiv preprint arXiv:1312.6114*, 2014. https://arxiv.org/abs/1312.6114

IN ORDER TO MOVE FROM DETERMINISTIC COMPRESSION TO PROBABILISTIC LATENT SPACES we have good reason to find ways to,

- derive a training objective that trades off faithful reconstruction against latent-space regularity.
- enable gradient-based optimisation through a stochastic sampling step that would otherwise block backpropagation.
- understand how structural choices (KL weight, discrete versus continuous codes, conditioning variables) shape the resulting latent space and what each variant buys us.

Hands-On Experience

IMAGINE you have a 2D latent space with a standard normal prior $p(z) = \mathcal{N}(0, I)$.

Sample $z_1 = (1, 0)$ and $z_2 = (-1, 0)$. Interpolate: $z_t = (1-t)\, z_1 + t\, z_2$ for $t \in [0, 1]$. Decode each z_t.

What do you expect the decoded outputs to look like along this path? Why might a standard autoencoder fail to produce a meaningful interpolation?

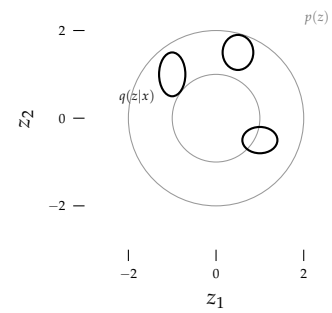


Figure 13: VAE latent space: each input encodes to a distribution (ellipses), regularised toward the prior (circles).

THE FUNDAMENTAL QUESTION is: can we turn the latent bottleneck into a proper probability distribution, so that every point in the latent space decodes to something meaningful and we can generate new data by sampling?

THE LEARNING OBJECTIVES of this lecture:

- Derive the Evidence Lower Bound and explain how its two terms trade off reconstruction against regularisation.
- Implement the reparameterisation trick to enable gradient-based training through a stochastic bottleneck.
- Distinguish VAE variants, β-VAE, VQ-VAE, and conditional VAE, by the structural assumption each changes.

From Autoencoders to VAEs

A STANDARD AUTOENCODER consists of an encoder $f_\phi \colon x \mapsto z$, a decoder $g_\theta \colon z \mapsto \hat{x}$, and a reconstruction loss $\mathcal{L} = \|x - g_\theta(f_\phi(x))\|^2$.

STANDARD AUTOENCODERS learn $x \to z \to \hat{x}$, minimising $\|x - \hat{x}\|^2$.

THREE PROBLEMS EMERGE once we look beyond reconstruction. The latent space may contain holes, regions where no training point maps, and decoding a z from such a region may produce nonsense. There is no principled way to sample new data from the latent space. And there is no measure of how uncertain a given representation is.

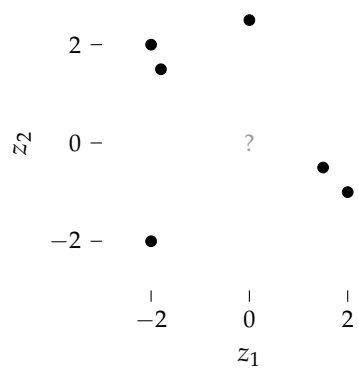


Figure 14: Autoencoder latent space. Six inputs map to six scattered points. The centre of the space contains no training encodings; decoding a z sampled from that region is undefined.

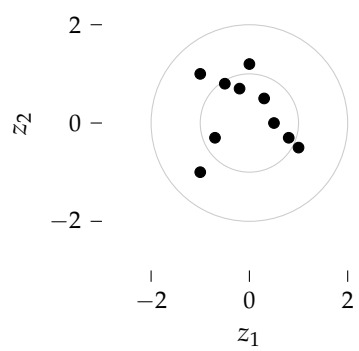


Figure 15: VAE latent space. The KL regularisation pushes encodings toward the standard normal prior (circles at 1σ and 2σ). Points fill the space; every region decodes to something meaningful.

The Evidence Lower Bound

WE WANT TO MODEL DATA WITH LATENT VARIABLES. The marginal likelihood of an observation x requires integrating over all possible latent codes:

THE GOAL is to maximise $\log p(x)$, but marginalising over z is intractable.

$$p_\theta(x) = \int p_\theta(x|z)\, p(z)\, dz$$

This integral is intractable for complex decoders. The posterior $p_\theta(z|x)$ is equally intractable.

VARIATIONAL INFERENCE sidesteps the intractable posterior by introducing a tractable approximation $q_\phi(z|x)$, parameterised by an encoder network.

$$\begin{aligned}
\log p_\theta(x) &= \log \int p_\theta(x|z)\, p(z)\, dz \\
&= \log \int \frac{q_\phi(z|x)}{q_\phi(z|x)}\, p_\theta(x|z)\, p(z)\, dz \\
&= \log \mathbb{E}_{q_\phi(z|x)} \left[\frac{p_\theta(x|z)\, p(z)}{q_\phi(z|x)} \right]
\end{aligned}$$

APPLYING JENSEN'S INEQUALITY, $\log \mathbb{E}[X] \geq \mathbb{E}[\log X]$, yields a lower bound on the log-evidence:

$$\log p_\theta(x) \geq \mathbb{E}_{q_\phi(z|x)} \left[\log p_\theta(x|z) + \log p(z) - \log q_\phi(z|x) \right]$$

REARRANGING GIVES THE ELBO:

$$\mathrm{ELBO}(\theta, \phi;\, x) = \underbrace{\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]}_{\text{Reconstruction}} - \underbrace{D_{\mathrm{KL}}\big(q_\phi(z|x) \,\|\, p(z)\big)}_{\text{Regularisation}}$$

ELBO stands for Evidence Lower BOund. Maximising the ELBO approximately maximises $\log p(x)$.

THE TWO TERMS TRADE OFF AGAINST EACH OTHER. The reconstruction term rewards an encoder and decoder pair that faithfully reconstructs x from a sampled z. The KL term penalises encoder distributions that stray far from the prior $p(z) = \mathcal{N}(0, I)$, ensuring the latent space remains regular.

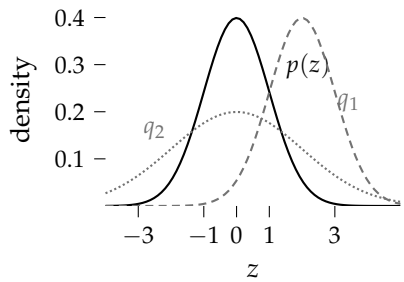


Figure 16: The KL penalty in action. $p(z) = \mathcal{N}(0,1)$ is the prior (solid). $q_1 = \mathcal{N}(2,1)$ (dashed) is shifted; $q_2 = \mathcal{N}(0,4)$ (dotted) is widened. Both deviations cost KL, but spreading mass away from the prior (q_2, D_{KL}=0.81) costs more than translating it (q_1, D_{KL}=0.50, evaluated in Exercise 2).

The Reparameterisation Trick

SAMPLING IS NOT DIFFERENTIABLE. The reparameterisation trick moves the stochasticity outside the computational graph.

TO OPTIMISE THE ELBO we need gradients through the expectation $\mathbb{E}_{q_\phi(z|x)}[\cdot]$. Sampling from $q_\phi(z|x)$ blocks gradient flow because the sample operation has no derivative with respect to ϕ.

THE FIX: instead of sampling $z \sim \mathcal{N}(\mu_\phi(x), \sigma^2_\phi(x))$ directly, reparameterise as

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

Now the randomness lives in ϵ, which does not depend on ϕ. Gradients flow through μ and σ as usual.

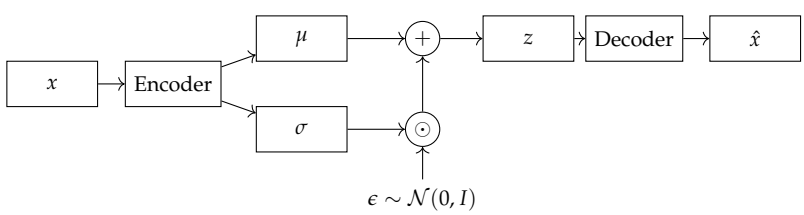


Figure 17: VAE architecture with reparameterisation trick: $z = \mu + \sigma \odot \epsilon$. Stochasticity enters through ϵ; all other arrows carry deterministic, differentiable operations.

VAE Training

THE ENCODER outputs $\mu_\phi(x)$ and $\log \sigma^2_\phi(x)$; the decoder outputs reconstruction parameters; the prior is $p(z) = \mathcal{N}(0, I)$.

FOR A GAUSSIAN DECODER WITH FIXED VARIANCE, the ELBO simplifies to a concrete loss:

$$\mathcal{L}(\theta, \phi; x) = \|x - \hat{x}\|^2 + D_{\mathrm{KL}}\big(q_\phi(z|x) \,\|\, p(z)\big)$$

THE KL TERM HAS A CLOSED FORM when both distributions are Gaussian:

$$D_{\mathrm{KL}}\big(\mathcal{N}(\mu,\sigma^2)\,\|\,\mathcal{N}(0,1)\big) = \tfrac{1}{2}\sum_j\big(\mu_j^2+\sigma_j^2-\log\sigma_j^2-1\big)$$

NUMERICAL STABILITY. In practice the encoder outputs $\log\sigma^2$ rather than σ directly, avoiding negative-variance issues and improving gradient behaviour.

Exploring the Latent Space

LATENT INTERPOLATION reveals the structure that the VAE has learned. Given two data points x_1 and x_2, encode them to $z_1 = \mu_\phi(x_1)$ and $z_2 = \mu_\phi(x_2)$, interpolate as $z_t = (1-t)\,z_1 + t\,z_2$, and decode each z_t to obtain $\hat{x}_t = g_\theta(z_t)$. Because the latent space is regularised, the decoded path transitions smoothly between the two inputs.

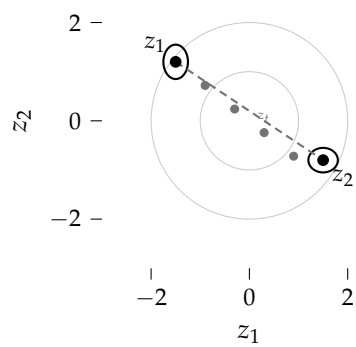


Figure 18: Latent interpolation. Two inputs encode to z_1 and z_2 (ellipses show the encoder uncertainty). The dashed line interpolates $z_t = (1-t)\,z_1 + t\,z_2$; each z_t is decoded to produce a smooth transition. Because the latent space is regularised, no point along the path falls into a hole.

GENERATION is equally straightforward: sample $z \sim \mathcal{N}(0, I)$ and decode $\hat{x} = g_\theta(z)$. The regularisation ensures that every region of the prior generates plausible data.

VAE Variants

β-VAE[3] increases the weight on the KL term:

$$\mathcal{L}_\beta = \text{Reconstruction} + \beta \cdot D_{\text{KL}}$$

With $\beta > 1$ the model is pressured to use the latent space efficiently, often leading to disentangled representations where each latent dimension corresponds to a single factor of variation.

β-VAE trades reconstruction fidelity for disentanglement by scaling the KL term.

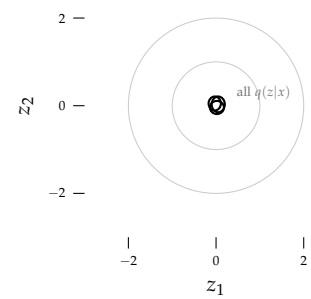


Figure 19: Posterior collapse. With high β, all encoder distributions collapse to the prior: $q(z|x) \approx \mathcal{N}(0, I)$ for every input. The latent code carries no information about x.

[3] I. Higgins, L. Matthey, A. Pal, C. Burgess, X. Glorot, M. Botvinick, S. Mohamed, and A. Lerchner. β-VAE: Learning basic visual concepts with a constrained variational framework. In *International Conference on Learning Representations (ICLR)*, 2017. https://scholar.google.com/scholar?q=Higgins+2017+beta-VAE+learning+basic+visual+concepts

VQ-VAE[4] replaces continuous latents with discrete codes from a learned codebook. Each encoder output z_e is snapped to the nearest codebook entry:

$$z_q = \arg\min_{e_k \in \text{codebook}} \|z_e - e_k\|$$

Discrete codes sidestep posterior collapse entirely and enable autoregressive priors that capture complex dependencies between code positions.

[4] A. van den Oord, O. Vinyals, and K. Kavukcuoglu. Neural discrete representation learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017. https://arxiv.org/abs/1711.00937

VQ-VAE enables powerful autoregressive priors, such as PixelCNN, over the discrete codes.

Conditional VAE[5] adds a conditioning variable c, such as a class label, to both encoder $q_\phi(z|x, c)$ and decoder $p_\theta(x|z, c)$. This enables controlled generation: sample z, specify c, and decode.

[5] K. Sohn, H. Lee, and X. Yan. Learning structured output representation using deep conditional generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 28, 2015. https://scholar.google.com/scholar?q=Sohn+Lee+Yan+2015+learning+structured+output+conditional+generative

Brief: Knowledge Distillation

Knowledge distillation[6] compresses a large teacher model into a smaller student by training the student to match the teacher's soft output distributions rather than hard labels.

Distillation transfers knowledge from a large teacher to a small student.

[6] G. Hinton, O. Vinyals, and J. Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015. https://arxiv.org/abs/1503.02531

Soft labels are produced by scaling the teacher's logits with a temperature T:

$$p_i^{(T)} = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

Higher T yields a softer distribution that reveals more about the teacher's learned structure.

The connection to VAEs is conceptual: both methods learn compressed representations that preserve essential information, one in the latent space, the other in the student's weights.

Examples & Exercises

COMPUTING BY HAND builds intuition for what the ELBO actually measures. Step through the arithmetic slowly.

ELBO DERIVATION. Starting from $\log p(x) = \log \int p(x,z)\,dz$, derive the ELBO. Work through it in four steps.

STEP 1: introduce $q(z|x)$ by multiplying and dividing inside the integral:

$$\begin{aligned}\log p(x) &= \log \int \frac{q(z|x)}{q(z|x)}\, p(x,z)\,dz \\ &= \log \mathbb{E}_{q(z|x)}\left[\frac{p(x,z)}{q(z|x)}\right]\end{aligned}$$

STEP 2: apply Jensen's inequality, $\log \mathbb{E}[X] \geq \mathbb{E}[\log X]$:

$$\log p(x) \geq \mathbb{E}_{q(z|x)}\left[\log \frac{p(x,z)}{q(z|x)}\right]$$

Complete Steps 3 and 4 yourself: split $p(x,z) = p(x|z)\,p(z)$ and rearrange to obtain the ELBO in reconstruction + KL form. Verify that you arrive at:

$$\text{ELBO} = \underbrace{\mathbb{E}_{q(z|x)}[\log p(x|z)]}_{\text{Reconstruction}} - \underbrace{D_{\text{KL}}\big(q(z|x) \,\|\, p(z)\big)}_{\text{Regularisation}}$$

WHY A LOWER BOUND? Jensen's inequality introduces a gap: $\log p(x) = \text{ELBO} + D_{\text{KL}}(q \| p(z|x))$. The gap is the KL divergence between the approximate and true posterior. Maximising the ELBO simultaneously improves reconstruction and pushes q toward the true posterior.

KL DIVERGENCE BY HAND. Compute $D_{\text{KL}}\big(\mathcal{N}(\mu,\sigma^2) \,\|\, \mathcal{N}(0,1)\big)$ for three settings. Use the closed form $D_{\text{KL}} = \frac{1}{2}(\mu^2 + \sigma^2 - \log \sigma^2 - 1)$.

SETTING (1): μ=0, σ=1 (the posterior matches the prior exactly).

$$\begin{aligned}D_{\text{KL}} &= \tfrac{1}{2}(0^2 + 1^2 - \log 1^2 - 1) \\ &= \tfrac{1}{2}(0 + 1 - 0 - 1) \\ &= 0\end{aligned}$$

As expected: when $q(z|x)$ exactly matches $p(z)$, the KL divergence is zero.

SETTING (II): μ=1, σ=1 (shifted mean, same spread).

$$\begin{aligned} D_{\mathrm{KL}} &= \tfrac{1}{2}(1^2 + 1^2 - \log 1^2 - 1) \\ &= \tfrac{1}{2}(1 + 1 - 0 - 1) \\ &= 0.5 \end{aligned}$$

The encoder shifted its mean away from zero. The prior penalises this: a shifted Gaussian wastes probability mass in regions the prior considers unlikely.

SETTING (III): μ=0, σ=2 (correct mean, too wide). Compute this yourself, then verify:

$$\begin{aligned} D_{\mathrm{KL}} &= \tfrac{1}{2}(0^2 + 2^2 - \log 2^2 - 1) \\ &= \tfrac{1}{2}(0 + 4 - 1.386 - 1) \\ &\approx 0.81 \end{aligned}$$

Setting	μ	σ	D_{KL}	**Deviation from prior**
(i)	0	1	0	None (exact match)
(ii)	1	1	0.50	Shifted mean
(iii)	0	2	0.81	Too wide

Table 4: KL divergence for three encoder settings. A wider posterior (iii) costs more than a shifted one (ii) because spreading probability mass away from the prior is more expensive than translating it.

REPARAMETERISATION. Why can we not directly backpropagate through $z \sim \mathcal{N}(\mu, \sigma^2)$? Write out the reparameterised form, then verify the gradients.

THE PROBLEM: sampling $z \sim \mathcal{N}(\mu, \sigma^2)$ is a stochastic operation with no gradient. The parameters μ and σ determine the distribution, but the sample itself has no differentiable relationship to them.

THE FIX: reparameterise as $z = \mu + \sigma\,\epsilon$ where $\epsilon \sim \mathcal{N}(0, 1)$. Now z is a deterministic function of μ, σ, and ϵ. The randomness lives in ϵ, which does not depend on the parameters.

NUMERICAL FORWARD PASS. Let μ=0.5, σ=1.2, and suppose $\epsilon = -$

0.3 is sampled:

$$\begin{aligned} z &= \mu + \sigma \cdot \epsilon \\ &= 0.5 + 1.2 \times (-0.3) \\ &= 0.5 - 0.36 \\ &= 0.14 \end{aligned}$$

VERIFY THE GRADIENTS:

$$\begin{aligned} \frac{\partial z}{\partial \mu} &= 1 \\ \frac{\partial z}{\partial \sigma} &= \epsilon \\ &= -0.3 \end{aligned}$$

Both are well-defined scalars. Without reparameterisation, these gradients would not exist: backpropagation would stop at the sampling step and the encoder would receive no learning signal.

ELBO EVALUATION BY THE NUMBERS. Combine the KL computation from the previous exercises with a concrete reconstruction to evaluate the full ELBO loss.

SETUP. Consider a 1D VAE with encoder output μ=1, σ=1 and input x=3. After sampling and decoding, the reconstruction is $\hat{x}$=2.7.

RECONSTRUCTION LOSS (squared error):

$$\begin{aligned} \mathcal{L}_{\text{recon}} &= (x - \hat{x})^2 \\ &= (3 - 2.7)^2 \\ &= 0.09 \end{aligned}$$

KL TERM (from Setting (ii) above):

$$D_{\text{KL}} = 0.50$$

TOTAL LOSS:

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{recon}} + D_{\text{KL}} \\ &= 0.09 + 0.50 \\ &= 0.59 \end{aligned}$$

Now compute the loss for a second configuration: μ=0, σ=1, x=3, $\hat{x}$=2.0. Which term dominates? What does this tell you about the reconstruction-regularisation balance?

THE TRADE-OFF IS VISIBLE. The reconstruction error is small (the decoder did a decent job), but the KL penalty is five times larger (the encoder strayed from the prior). A well-trained VAE balances both terms.

SECOND CONFIGURATION:

$$\begin{aligned}\mathcal{L}_{\text{recon}} &= (3-2.0)^2 \\ &= 1.00 \\ D_{\text{KL}} &= 0 \quad \text{(Setting (i): matches prior exactly)} \\ \mathcal{L} &= 1.00 + 0 \\ &= 1.00\end{aligned}$$

The encoder obeys the prior perfectly (D_{KL}=0), but the reconstruction suffers. The first configuration ($\mathcal{L}$=0.59) achieves a better total loss by accepting some KL cost in exchange for much better reconstruction.

POSTERIOR COLLAPSE. Consider a VAE trained with β=10 in the loss $\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta \cdot D_{\text{KL}}$. After training, you observe that $D_{\text{KL}} \approx 0$ for every input, and the reconstructions are blurry averages. Diagnose what has happened.

THE DIAGNOSIS: with β=10, the KL penalty is ten times more expensive than reconstruction. The cheapest way to minimise the loss is to set $q(z|x) = p(z) = \mathcal{N}(0, I)$ for every input. The encoder ignores x entirely: $\mu \to 0$, $\sigma \to 1$.

THE LATENT CODE CARRIES NO INFORMATION. Sampling $z \sim \mathcal{N}(0, I)$ produces the same distribution whether the input is a "3" or an "8". The decoder cannot distinguish inputs, so it reconstructs the average digit: a blurry blob.

VERIFY. If μ=0 and σ=1 for all inputs, then D_{KL}=0 (Setting (i) from the KL exercise). The KL term contributes nothing. The decoder receives the same distribution regardless of x, so it learns to output the mean of the training data.

TWO REMEDIES.

- **KL annealing:** start training with β=0 (autoencoder mode), then slowly raise β toward 1. The encoder first learns to use the latent space, then gradually accepts the regularisation.
- **Free bits:** allow each latent dimension to use at least λ nats of KL before penalising. This guarantees the encoder encodes at least some information per dimension.

AN MNIST DIGIT DATASET TO EXPLORE. Train a VAE with a 2D latent space on MNIST. Scatter-plot the latent encodings colour-coded

CODE will be provided as a Python notebook. Use it as a starting point, break things, and observe what changes.

by digit class. Sample a grid of points from the latent space, decode them, and display the resulting digits. Interpolate between two digits in latent space and display the decoded path.

WHAT TO OBSERVE. Digit classes should form overlapping but distinguishable clusters in the 2D latent space. The grid of decoded samples should show smooth transitions between digit types. Increasing β should sharpen the cluster boundaries but blur the reconstructions.

Self-Reflection and Recap

SELF-REFLECTION questions to guide your thinking:

- Why does the ELBO contain two terms, and what does each one enforce?
- What happens if you remove the KL term entirely and train with reconstruction loss alone?
- Why does the reparameterisation trick work, and what mathematical property of the Gaussian does it exploit?
- What is posterior collapse, and under what conditions does it occur?
- How does β-VAE encourage disentanglement, and what is sacrificed in return?
- Why does VQ-VAE avoid posterior collapse, and what new challenge does discrete quantisation introduce?
- When would you choose a conditional VAE over an unconditional one?
- How does the VAE latent space differ from the PCA subspace of the previous chapter, and what does the difference buy you?

RECAP of Key Concepts:

- VAEs learn probabilistic latent representations by maximising the ELBO, which trades off reconstruction fidelity against regularisation toward the prior.
- The reparameterisation trick enables gradient-based training by expressing the stochastic sample as a deterministic function of the parameters plus independent noise.

- The closed-form KL for Gaussians makes training efficient; in practice the encoder outputs $\log \sigma^2$ for numerical stability.
- β-VAE increases the KL weight to encourage disentanglement at the cost of reconstruction quality; VQ-VAE replaces continuous latents with discrete codes to enable autoregressive priors.
- Knowledge distillation is a complementary compression strategy: it transfers a teacher's soft predictions into a smaller student network.

A VAE TRAINED ON COFFEE AROMA PROFILES will reconstruct familiar profiles faithfully and assign them high ELBO. A contaminated sample, one the model has never seen, will reconstruct poorly and land in a low-density region of the latent space. The model signals that something is off, but it cannot quantify how uncertain it is, or decide what to do about it.

THE ROOT OF VAE BLINDNESS. A generative model can tell you what the data looks like, but not how much to trust what it says about data it has never seen.

UNCERTAINTY ESTIMATION fills this gap. The next chapter develops tools for quantifying what a model does and does not know, distinguishing irreducible noise from reducible ignorance, and showing how uncertainty estimates enable anomaly detection, out-of-distribution flagging, and graceful degradation in deployed systems.

TEASER. A VAE assigns a reconstruction error to every input. Is a high error sufficient to declare an input anomalous, or do we need a principled measure of what the model does not know?

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